Theory of Statistics

Axiomatic Foundations, Proofs and Computational Applications

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Preface

This book aims to present the core ideas of mathematical statistics as a single whole, starting from measure-theoretic probability and continuing through point estimation, hypothesis testing, interval estimation and Bayesian inference. The final part unifies these topics within decision theory (minimaxity and admissibility) and extends them to three topics that arise often in practice: nonparametric methods, multiple testing and the false discovery rate, and the EM algorithm together with generalized linear models. Every chapter follows the same structure:

1. **Introduction and motivation:** the question the chapter sets out to answer.
2. **Definitions:** precise, axiomatic statements.
3. **Theorems and proofs:** the main results, with their assumptions stated explicitly.
4. **Worked examples:** what the theory says for the classical distributions.
5. **Computational applications:** Python code that checks the theoretical results by Monte Carlo simulation, together with the actual output of that code.
6. **Comments:** the limits of the results, common mistakes and links to other chapters.
7. **Chapter summary:** a box collecting the main results with their theorem numbers, and a note pointing to the corresponding chapters of standard textbooks.
8. **Exercises:** for consolidating the material.

Appendices. Appendix **??** at the end of the book is a reference table of the densities, means, variances and MGFs of common distributions, the relations between them and their conjugate priors. Appendix A gives solutions to all the exercises; the link at the end of each exercise leads to its solution, and the heading of each solution leads back to the exercise. The *Index* at the very end lists the pages on which the main concepts appear; the page on which a concept is defined is set in bold.

About the code. All code is written in Python 3 with `numpy`, `scipy` and `matplotlib`. Each code block is a separate file in the `code/` folder of the book (the file name is printed above the code box) and runs on its own; all of them can be run with `python3 code/run_all.py`. The simulations use numpy's recommended random number interface:

```
rng = np.random.default_rng(42)
```

Because the seed is fixed, the results are reproducible. The "Output" boxes directly below the code blocks were produced by actually running the code. Throughout the book the Gamma and exponential distributions are written with a *rate* parameter (see Notation), whereas `numpy` and `scipy` parametrise them by the *scale* $= 1/\text{rate}$. Expressions such as `scale=1/beta` in the code perform this conversion.

Notation

Symbol	**Meaning**
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space
$\mathcal{B}(\mathbb{R})$	Borel σ-algebra on $\mathbb{R}$
F_X, f_X, p_X	Distribution function (CDF), density (pdf), probability mass function (pmf)
$\mathbb{E}$, Var, Cov	Expectation, variance, covariance
$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$	Independent and identically distributed sample
$\bar{X}_n$, S_n^2	Sample mean, unbiased sample variance (divided by $n-1$)
$X_{(1)} \le \cdots \le X_{(n)}$	Order statistics
$\xrightarrow{P}$, $\xrightarrow{d}$, $\xrightarrow{\text{a.s.}}$, $\xrightarrow{L^p}$	Convergence in probability, in distribution, almost surely, in L^p
$L(\theta; \mathbf{x})$, $\ell(\theta; \mathbf{x})$	Likelihood and log-likelihood function
$I_1(\theta)$, $I_n(\theta)$	Fisher information for one and for n observations
Φ, φ, z_α	Standard normal CDF, pdf and upper α quantile: $\Phi(z_\alpha) = 1-\alpha$
$\chi^2_{k,\alpha}$, $t_{k,\alpha}$	Upper α quantiles of the chi-square and t distributions
Gamma(α, β)	Shape α, *rate* β: $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, mean α/β
Exp(λ)	Rate λ: Gamma$(1, \lambda)$, mean $1/\lambda$; also $\chi^2_k = \text{Gamma}(k/2, \frac{1}{2})$
$\mathbb{I}_A$	Indicator function of the event A
$\perp\!\!\!\perp$	Independence

Contents

Preface **i**

I Foundations of Probability Theory **1**

1 Probability Spaces and Kolmogorov's Axioms **2**
1.1 Definitions 2
1.2 Main Theorems and Proofs 3
1.3 Worked Examples 4
1.4 Computational Applications 5
1.4.1 Verifying Continuity by Monte Carlo 5
1.4.2 The Birthday Problem: Exact Value and Simulation 6
1.4.3 Bayes' Formula: Simulating a Diagnostic Test 6
1.5 Comments and Notes 6
1.6 Chapter Summary 7
1.7 Exercises 7

II Appendices **8**

A Solutions to the Exercises **9**
A.1 Probability Spaces 9

Bibliography **11**

Part I

Foundations of Probability Theory

Chapter 1

Probability Spaces and Kolmogorov's Axioms

Probability theory is the mathematics of assigning numerical weights, in a consistent way, to the outcomes of experiments that involve uncertainty. The word "consistent" is crucial: unless we specify in advance which sets may receive a probability (the σ-algebra) and which rules the assignment must obey (Kolmogorov's axioms), arguments that look intuitively correct can lead to contradictions. In this chapter we lay the foundation on which the rest of the book is built.

1.1 Definitions

Definition 1.1 (Sample Space and Event)**.** The set of all possible outcomes of a random experiment is called the *sample space* and is denoted by Ω. Its elements $\omega \in \Omega$ are called *outcomes* (or elementary events), and the subsets $A \subseteq \Omega$ to which a probability can be assigned are called *events*.

Definition 1.2 (σ-Algebra)**.** A family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ of subsets of the sample space Ω is a σ-algebra if

1. $\Omega \in \mathcal{F}$,
2. $A \in \mathcal{F} \implies A^c = \Omega \setminus A \in \mathcal{F}$,
3. $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a *measurable space*. If the third condition is required only for finite unions, $\mathcal{F}$ is called an *algebra* (or field).

Remark 1.3. By De Morgan's law $\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c$, a σ-algebra is also closed under countable intersections, under differences ($A \setminus B = A \cap B^c$), and contains $\emptyset = \Omega^c$.

Definition 1.4 (Generated σ-Algebra)**.** Let $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ be any family of sets. The intersection of all σ-algebras containing $\mathcal{C}$,

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{G} : \mathcal{G} \text{ is a } \sigma\text{-algebra and } \mathcal{C} \subseteq \mathcal{G}\},$$

is again a σ-algebra, and it is the *smallest* σ-algebra containing $\mathcal{C}$. (The intersection is over a non-empty family, since $\mathcal{P}(\Omega)$ always belongs to it.)

Definition 1.5 (Kolmogorov's Axioms of Probability [43])**.** A function $\mathbb{P} : \mathcal{F} \to [0, 1]$ defined on a measurable space $(\Omega, \mathcal{F})$ is a *probability measure* if it satisfies:

1. (Non-negativity) $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{F}$.
2. (Normalisation) $\mathbb{P}(\Omega) = 1$.
3. (Countable additivity) For every sequence $\{A_i\}_{i=1}^{\infty}$ of pairwise disjoint events, $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a *probability space*.

Definition 1.6 (Conditional Probability)**.** For $\mathbb{P}(B) > 0$, the conditional probability of A given B is defined by

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For fixed B, the map $A \mapsto \mathbb{P}(A \mid B)$ is a new probability measure on $(\Omega, \mathcal{F})$.

Definition 1.7 (Independence)**.** Events $A, B \in \mathcal{F}$ are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. A family $\{A_i\}_{i \in \mathcal{I}}$ is *mutually independent* if

$$\mathbb{P}\Big(\bigcap_{j \in J} A_j\Big) = \prod_{j \in J} \mathbb{P}(A_j)$$

for every finite $J \subseteq \mathcal{I}$. If the condition is required only for $|J| = 2$, the family is *pairwise independent*.

Definition 1.8 (Limit Superior and Limit Inferior of a Sequence of Events)**.** For a sequence of events $\{A_n\}$,

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \{A_n \text{ occurs infinitely often}\}, \qquad \liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k.$$

$\liminf A_n$ is the event "from some index on, all the A_n occur". Both belong to $\mathcal{F}$.

1.2 Main Theorems and Proofs

Proposition 1.9 (Basic Properties of a Probability Measure)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A, B \in \mathcal{F}$.*

1. $\mathbb{P}(\emptyset) = 0$.
2. *(Finite additivity) If $A_1, \dots, A_n$ are pairwise disjoint, then $\mathbb{P}(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i)$.*
3. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
4. *(Monotonicity) $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.*
5. *(Inclusion–exclusion) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.*
6. *(Boole's inequality) For every $\{A_i\} \subseteq \mathcal{F}$, $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.*

Proof. (1) Applying Axiom 3 to the disjoint sequence $A_1 = \Omega$, $A_i = \emptyset$ $(i \geq 2)$ gives $1 = 1 + \sum_{i \geq 2} \mathbb{P}(\emptyset)$; a sum of non-negative terms equal to zero forces $\mathbb{P}(\emptyset) = 0$.

(2) Take $A_{n+1} = A_{n+2} = \cdots = \emptyset$ and use Axiom 3 together with (1).

(3) From the disjoint union $\Omega = A \cup A^c$ we get $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$.

(4) $B = A \cup (B \setminus A)$ is a disjoint union; hence $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.

(5) From the disjoint unions $A \cup B = A \cup (B \setminus A)$ and $B = (A \cap B) \cup (B \setminus A)$ we obtain $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

(6) The sets $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{i<n} A_i$ are pairwise disjoint, $B_n \subseteq A_n$ and $\bigcup B_n = \bigcup A_n$. By Axiom 3 and monotonicity, $\mathbb{P}(\bigcup A_n) = \sum \mathbb{P}(B_n) \leq \sum \mathbb{P}(A_n)$. $\square$

Theorem 1.10 (Continuity of Probability)**.** *If $A_1 \subseteq A_2 \subseteq \dots$ is an increasing sequence of events and $A = \bigcup_{n=1}^{\infty} A_n$, then*

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A).$$

Proof. Define disjoint rings: $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1} \in \mathcal{F}$ for $n \geq 2$. The sets are pairwise disjoint, $\bigcup_{i=1}^{n} C_i = A_n$ and $\bigcup_{i=1}^{\infty} C_i = A$. By Kolmogorov's Axiom 3 and finite additivity,

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} C_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(C_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(C_i) = \lim_{n\to\infty} \mathbb{P}(A_n). \qquad \square$$

Corollary 1.11 (Continuity from Above)**.** *If $A_1 \supseteq A_2 \supseteq \dots$ is a decreasing sequence and $A = \bigcap_{n=1}^{\infty} A_n$, then $\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.*

Proof. The sequence A_n^c is increasing, and by De Morgan $\bigcup_n A_n^c = A^c$. By Theorem 1.10, $1 - \mathbb{P}(A_n) = \mathbb{P}(A_n^c) \to \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. $\square$

Remark 1.12. Theorem 1.10 is in fact *equivalent* to Axiom 3: every finitely additive set function that is continuous from below is countably additive. Continuity is thus the "analytic face" of countable additivity. In Chapter **??**, the right-continuity of the CDF will follow directly from Corollary 1.11.

Theorem 1.13 (Law of Total Probability and Bayes' Formula)**.** *Let $\{B_i\}_{i\geq 1}$ be a countable partition of Ω (pairwise disjoint with $\bigcup B_i = \Omega$) with $\mathbb{P}(B_i) > 0$. For every $A \in \mathcal{F}$,*

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i), \qquad \mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)} \quad (\mathbb{P}(A) > 0).$$

Proof. $A = \bigcup_i (A \cap B_i)$ is a disjoint union; by Axiom 3 and the definition of conditional probability, $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$. Bayes' formula follows by writing the numerator and denominator of $\mathbb{P}(B_j \mid A) = \mathbb{P}(A \cap B_j)/\mathbb{P}(A)$ in this way. $\square$

Theorem 1.14 (Borel–Cantelli Lemmas; see [22, Section 2.3])**.** *Let $\{A_n\}$ be a sequence of events.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(\limsup A_n) = 0$.*
2. *If the $\{A_n\}$ are mutually independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(\limsup A_n) = 1$.*

Proof. (1) For every n, $\limsup A_k \subseteq \bigcup_{k\geq n} A_k$; by Boole's inequality $\mathbb{P}(\limsup A_k) \leq \sum_{k\geq n} \mathbb{P}(A_k)$. The right-hand side is the tail of a convergent series and tends to zero as $n \to \infty$.

(2) Since $(\limsup A_n)^c = \bigcup_n \bigcap_{k\geq n} A_k^c$, it suffices to show that $\mathbb{P}(\bigcap_{k\geq n} A_k^c) = 0$ for every n. By independence and the inequality $1 - x \leq e^{-x}$, for every $m > n$,

$$\mathbb{P}\Big(\bigcap_{k=n}^{m} A_k^c\Big) = \prod_{k=n}^{m} (1 - \mathbb{P}(A_k)) \leq \exp\Big(-\sum_{k=n}^{m} \mathbb{P}(A_k)\Big) \xrightarrow{m\to\infty} 0.$$

Corollary 1.11 then gives $\mathbb{P}(\bigcap_{k\geq n} A_k^c) = 0$. $\square$

1.3 Worked Examples

Example 1.15 (A Finite σ-Algebra)**.** For a roll of a die let $\Omega = \{1, \dots, 6\}$. If we are interested only in the information "odd or even?", then $\mathcal{F} = \{\emptyset, \{1,3,5\}, \{2,4,6\}, \Omega\}$ is a σ-algebra. This $\mathcal{F}$ does not contain the event $\{1\}$, so at this level of information no probability can be assigned to the event "a 1 was rolled". Interpreting a σ-algebra as a body of *information* is the basis of the conditional expectation in Chapter **??**.

Example 1.16 (An Algebra That Is Not a σ-Algebra)**.** Let $\Omega = \mathbb{N}$ and $\mathcal{A} = \{A \subseteq \mathbb{N} : A \text{ is finite or } A^c \text{ is finite}\}$. $\mathcal{A}$ is closed under complements and finite unions, so it is an algebra. However, the set of even numbers, which is the union of the finite sets $A_i = \{2i\}$, is neither finite nor co-finite; hence $\mathcal{A}$ is not a σ-algebra.

Example 1.17 (The Birthday Problem)**.** Suppose the birthdays in a group of n people are independent and uniformly distributed over 365 days. By the complement rule, the probability that at least two people share a birthday is

$$p_n = 1 - \prod_{k=0}^{n-1} \frac{365 - k}{365}.$$

The approximation $1 - x \approx e^{-x}$ gives $p_n \approx 1 - \exp\{-n(n-1)/730\}$; for $n = 23$, $p_{23} \approx 0.507$. This counter-intuitive result is explained by the existence of $\binom{23}{2} = 253$ distinct pairs of people.

Example 1.18 (A Medical Diagnostic Test — Bayes' Formula)**.** Let the prevalence of a disease be $\mathbb{P}(D) = 0.01$, the sensitivity of the test $\mathbb{P}(+ \mid D) = 0.99$ and its specificity $\mathbb{P}(- \mid D^c) = 0.95$. By Theorem 1.13,

$$\mathbb{P}(D \mid +) = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0099}{0.0594} \approx 0.167.$$

A person with a positive result is actually ill with probability only 16.7%. A low prior probability (prevalence) lowers the positive predictive value of even a highly accurate test. This is the simplest instance of the discussion of "the role of prior information" in Chapter **??**.

Example 1.19 (Events That Are Pairwise but Not Mutually Independent)**.** Toss two fair coins: $A = \{\text{first toss H}\}$, $B = \{\text{second toss H}\}$, $C = \{\text{both tosses equal}\}$. Since $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/2$ and every pairwise intersection has probability $1/4$, the events are pairwise independent. However, $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\{HH\}) = 1/4 \neq 1/8 = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. Mutual independence is therefore a strictly stronger condition than pairwise independence.

Example 1.20 (An Application of Borel–Cantelli)**.** If the events A_n are independent with $\mathbb{P}(A_n) = 1/n^2$, then since $\sum 1/n^2 = \pi^2/6 < \infty$, with probability 1 only finitely many of the A_n occur. If $\mathbb{P}(A_n) = 1/n$, then $\sum 1/n = \infty$ and, under independence, the A_n occur infinitely often with probability 1. The difference between the two sequences decides whether "rare" events eventually die out.

1.4 Computational Applications

1.4.1 Verifying Continuity by Monte Carlo

For $\Omega = [0,1]$ with Lebesgue measure $\mathbb{P}$ and the increasing sequence $A_n = [0, 1 - 1/n]$, we have $A = \bigcup A_n = [0,1)$ and $\mathbb{P}(A) = 1$. By Theorem 1.10, $\mathbb{P}(A_n) = 1 - 1/n \to 1$.

code/ch01/c01_1_continuity_monte_carlo.py

```python
import numpy as np

rng = np.random.default_rng(42)
N_SAMPLES = 500_000
omega_samples = rng.uniform(0.0, 1.0, size=N_SAMPLES)

for n in [1, 5, 10, 25, 50]:
    threshold = 1.0 - (1.0 / n)
    p_sim = np.mean(omega_samples <= threshold)
    print(f"n = {n:2d} | Theoretical P(A_n): {threshold:.4f} | Monte Carlo: {p_sim:.4f}")
```

Output

```
n =  1 | Theoretical P(A_n): 0.0000 | Monte Carlo: 0.0000
n =  5 | Theoretical P(A_n): 0.8000 | Monte Carlo: 0.8005
n = 10 | Theoretical P(A_n): 0.9000 | Monte Carlo: 0.9000
n = 25 | Theoretical P(A_n): 0.9600 | Monte Carlo: 0.9601
n = 50 | Theoretical P(A_n): 0.9800 | Monte Carlo: 0.9799
```

1.4.2 The Birthday Problem: Exact Value and Simulation

code/ch01/c01_2_birthday_problem.py

```python
import numpy as np

rng = np.random.default_rng(42)
def exact(n):
    return 1.0 - np.prod((365 - np.arange(n)) / 365)

R = 100_000
for n in [10, 23, 30, 50]:
    days = rng.integers(0, 365, size=(R, n))
    days.sort(axis=1)
    collision = np.any(np.diff(days, axis=1) == 0, axis=1)
    approx = 1 - np.exp(-n * (n - 1) / 730)
    print(f"n={n:2d} | exact={exact(n):.4f} | exponential approx.={approx:.4f} | MC={
    collision.mean():.4f}")
```

Output

```
n=10 | exact=0.1169 | exponential approx.=0.1160 | MC=0.1179
n=23 | exact=0.5073 | exponential approx.=0.5000 | MC=0.5067
n=30 | exact=0.7063 | exponential approx.=0.6963 | MC=0.7086
n=50 | exact=0.9704 | exponential approx.=0.9651 | MC=0.9707
```

1.4.3 Bayes' Formula: Simulating a Diagnostic Test

code/ch01/c01_3_bayes_diagnostic_test.py

```python
import numpy as np

rng = np.random.default_rng(42)
N = 2_000_000
ill = rng.random(N) < 0.01
positive = np.where(ill, rng.random(N) < 0.99, rng.random(N) < 0.05)
print(f"P(D | +) theory    : {0.0099/0.0594:.4f}")
print(f"P(D | +) simulation: {ill[positive].mean():.4f}")
print(f"P(+)     theory    : {0.0594:.4f} | simulation: {positive.mean():.4f}")
```

Output

```
P(D | +) theory    : 0.1667
P(D | +) simulation: 0.1664
P(+)     theory    : 0.0594 | simulation: 0.0592
```

1.5 Comments and Notes

Remark 1.21 (Why Not Assign a Probability to Every Subset?). It is impossible to assign a translation-invariant, countably additive "length" to *all* subsets of $\Omega = [0,1]$ (the Vitali set; this requires the axiom of choice). We therefore define probability only on a σ-algebra of "sufficiently regular" sets. Every set met in practice (intervals, their countable unions and intersections, and so on) belongs to the Borel σ-algebra.

Remark 1.22 (Probability Zero Is Not Impossibility). In the uniform example above, $\mathbb{P}(\{\omega\}) = 0$ for every single point $\{\omega\}$, yet the outcome of the experiment is always some point. This is why events of probability 1 are said to occur "almost surely" (a.s.).

Remark 1.23 (Monte Carlo Error). When a probability p is estimated by the proportion in N independent trials, the standard error is $\sqrt{p(1-p)/N}$. For $N = 500\,000$ and $p = 0.5$ this is about 0.0007; the differences between the theoretical and simulated values in the outputs above are of this order. The proportion is guaranteed to converge by the Law of Large Numbers in Chapter **??**.

1.6 Chapter Summary

- A probability model is a triple $(\Omega, \mathcal{F}, \mathbb{P})$: the events form a σ-algebra (Definition 1.2) and $\mathbb{P}$ satisfies Kolmogorov's axioms, in particular countable additivity (Definition 1.5).
- The complement rule, monotonicity, inclusion–exclusion and Boole's inequality are derived from the axioms (Proposition 1.9).
- A probability measure is continuous along monotone sequences of events (Theorem 1.10). Being able to interchange limits and probability underlies all later convergence results.
- Conditional probability, the law of total probability and Bayes' formula (Theorem 1.13) are the probabilistic counterpart of Bayesian inference (Chapter **??**). Mutual independence is stronger than pairwise independence.
- Borel–Cantelli lemmas (Theorem 1.14): if $\sum \mathbb{P}(A_n) < \infty$, the A_n occur infinitely often with probability 0; for independent events with $\sum \mathbb{P}(A_n) = \infty$ this probability is 1. They are the main tool in proofs of almost sure convergence.

Further reading. For measure-theoretic foundations see [22, Chapter 1] and [9, Sections 1–4]; for a more introductory treatment see [16, Chapter 1] and [81, Chapter 1].

1.7 Exercises

Exercise 1.1. Show that if $\mathcal{F}_1$ and $\mathcal{F}_2$ are σ-algebras, then so is $\mathcal{F}_1 \cap \mathcal{F}_2$, but that $\mathcal{F}_1 \cup \mathcal{F}_2$ in general is not, by giving an example. → Solution

Exercise 1.2. Prove the inclusion–exclusion formula for three events and obtain the general formula for n events by induction. Use it to show that when n letters are placed in envelopes at random, the probability that no letter goes into its own envelope converges to e^{-1} as $n \to \infty$. → Solution

Exercise 1.3. Prove the Bonferroni inequality $\mathbb{P}(\bigcap_{i=1}^{n} A_i) \geq 1 - \sum_{i=1}^{n} \mathbb{P}(A_i^c)$. Discuss its use in multiple hypothesis testing (Chapter **??**). → Solution

Exercise 1.4. Show that if A and B are independent, then so are A and B^c, and A^c and B^c. → Solution

Exercise 1.5. Write Python code that plots $\mathbb{P}(D \mid +)$ in Example 1.18 as the prevalence $\mathbb{P}(D)$ varies over $[0.001, 0.5]$. Find analytically the prevalence threshold above which the positive predictive value exceeds 50%. → Solution

Part II

Appendices

Appendix A

Solutions to the Exercises

This appendix contains solutions to all the exercises in the book. The heading of each solution links to the exercise, and the "→ Solution" link at the end of each exercise leads here. Where an exercise asks for numerical values or a simulation, the values given were obtained by actual computation; simulation results may differ in the last digits depending on the random seed. The same libraries as in the chapters (`numpy`, `scipy`) are used; short code is given only where it is part of the solution.

The solutions are kept as short as possible, but each step names the result in the book on which it rests. When the claim of an exercise holds only under certain conditions (e.g. Exercise **??**), the solution states those conditions explicitly.

A.1 Probability Spaces

Chapter 1.

Exercise 1.1. Let $\mathcal{F} = \mathcal{F}_1 \cap \mathcal{F}_2$. Since Ω is in both, $\Omega \in \mathcal{F}$. If $A \in \mathcal{F}$, then A belongs to both families, and since both are closed under complements, $A^c \in \mathcal{F}_1 \cap \mathcal{F}_2$. The same argument works for countable unions. (The same proof works for the intersection of arbitrarily many σ-algebras; the definition of the generated σ-algebra rests on this.) Counterexample for the union: $\Omega = \{1,2,3\}$, $\mathcal{F}_1 = \{\emptyset, \{1\}, \{2,3\}, \Omega\}$, $\mathcal{F}_2 = \{\emptyset, \{2\}, \{1,3\}, \Omega\}$. Then $\{1\}, \{2\} \in \mathcal{F}_1 \cup \mathcal{F}_2$, but $\{1\} \cup \{2\} = \{1,2\}$ is in neither; the union is not closed under unions.

Exercise 1.2. Applying the two-event formula to $A \cup B$ and C and using distributivity gives

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(AB) - \mathbb{P}(AC) - \mathbb{P}(BC) + \mathbb{P}(ABC)$$

($AB = A \cap B$). The general formula

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{i_1 < \cdots < i_k} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$

follows by induction: for $B = \bigcup_{i \le n} A_i$, $\mathbb{P}(B \cup A_{n+1}) = \mathbb{P}(B) + \mathbb{P}(A_{n+1}) - \mathbb{P}\big(\bigcup_{i \le n}(A_i \cap A_{n+1})\big)$; apply the induction hypothesis to the two unions on the right and collect terms. *Matching problem:* let $A_i = \{\text{letter } i \text{ is in the right envelope}\}$. The probability that k given letters are in the right place is $(n-k)!/n!$, and there are $\binom{n}{k}$ such choices; the inner sum equals $1/k!$. The probability that no letter is in the right place is

$$1 - \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k!} = \sum_{k=0}^{n} \frac{(-1)^k}{k!} \xrightarrow{n \to \infty} e^{-1} \approx 0.368.$$

Convergence is very fast: for $n = 6$ the difference is below 10^{-4}.

Exercise 1.3. By De Morgan and Boole's inequality, $\mathbb{P}(\bigcap A_i) = 1 - \mathbb{P}(\bigcup A_i^c) \geq 1 - \sum \mathbb{P}(A_i^c)$. *Multiple testing:* for m true null hypotheses let $A_i = \{\text{test } i \text{ does not reject falsely}\}$. If each test is carried out at level α/m, then $\mathbb{P}(A_i^c) \leq \alpha/m$, and the probability of no false rejection at all is $\geq 1 - \alpha$. This is the Bonferroni correction (see Exercise **??**); it holds whatever the dependence between the tests, but it is conservative when the tests are positively dependent.

Exercise 1.4. $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)\mathbb{P}(B^c)$, so $A \perp\!\!\!\perp B^c$. Applying this result again to the pair (B^c, A) gives $B^c \perp\!\!\!\perp A^c$.

Exercise 1.5. With prevalence π the positive predictive value is $\mathrm{PPV}(\pi) = \frac{0.99\pi}{0.99\pi + 0.05(1-\pi)}$. Then $\mathrm{PPV} > \frac{1}{2} \iff 0.99\pi > 0.05(1-\pi) \iff \pi > \frac{0.05}{1.04} \approx 0.0481$. So unless the prevalence exceeds about 4.8%, most positive results are false positives. For the plot:

```
import numpy as np, matplotlib.pyplot as plt
pi = np.linspace(0.001, 0.5, 500)
ppv = 0.99*pi / (0.99*pi + 0.05*(1 - pi))
plt.semilogx(pi, ppv); plt.axhline(0.5, ls=":"); plt.axvline(0.05/1.04, ls=":")
plt.xlabel("prevalence"); plt.ylabel("P(D | +)"); plt.show()
```

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